IB Economics - Topic 2.2: Teacher and AI Grading Guide

Supply | 50 paired questions

Scope and assessment status

50 original paired questions: (a) 10 marks, (b) 15 marks. Questions 1-40 assess shared SL/HL content and questions 41-50 assess the HL-only assumptions underlying the law of supply. Difficulty: 10 Easy, 25 Medium and 15 Hard, independently of level.

The current IB Economics guide, first assessment 2022, Topic 2.2 governs scope: law and curve of supply, individual and market supply, costs of factors of production, joint and competitive supply, indirect taxes and subsidies, future price expectations, technology, number of firms, and movements and shifts. Diminishing marginal returns and increasing marginal costs are HL only. No price-elasticity calculation, equilibrium, tax incidence, welfare analysis or producer surplus is required.

The uploaded Tragakes textbook, Economics for the IB Diploma (second edition, 2012), Chapter 2 section 2.3, printed pp. 26-29, checks these mechanisms. Its older numbering and additional special cases do not override the current IB scope. In particular, the current outline specifies indirect taxes, so questions do not require its profit-tax discussion or its special vertical-supply case.

These 15-mark evaluation questions are adapted Paper 1-style practice on a narrow topic. Related mechanisms recur from different analytical angles; questions are alternative practice, not 50 independent syllabus outcomes. This is original educational guidance, not official IB marking material.

Coverage blueprint

Law of supply: 1-4

Individual and market supply: 5-8

Factor costs: 9-13

Competitive supply: 14-17

Joint supply: 18-20

Indirect taxes and subsidies: 21-24

Expectations: 25-28

Technology: 29-32

Number of firms: 33-36

Movements versus shifts: 37-40

HL law assumptions: 41-50

Marking principles

Award a best-fit mark for each part. These indicative explanations identify valid reasoning and misconceptions, not a marks-per-keyword checklist. Credit other accurate routes that answer the actual question.

Part (a): 10 marks

Practice bands: 0; 1-2 limited; 3-4 mainly descriptive; 5-6 partly developed; 7-8 clear analysis; 9-10 complete, accurate, well-integrated explanation. Consider definitions, causation, and required diagram together. These are abridged practice descriptors, not official IB markbands.

Part (b): 15 marks

Practice bands: 0; 1-3 very limited; 4-6 superficial; 7-9 developed but weakly balanced; 10-12 balanced evaluation; 13-15 sustained application, evaluation and justified judgment. A case must be linked to the mechanism, not merely named.

Diagrams and AI grading

For requested diagrams, check P vertical, Q supplied horizontal, upward S0, correct direction and distinction between an own-price movement on S and a non-price shift of whole S, and an explanation of points or fixed-price comparison. A supply-only diagram does not determine market equilibrium or actual sales. If the image is unclear, flag the uncertain feature instead of assuming accuracy. Accept economically equivalent presentations.

For each response record (a)/(b) marks and band evidence, conceptual errors, diagram assessment if required, quality of real-world application, and one actionable improvement for each part. No particular suggested case or policy conclusion is mandatory. Source cases below are factual anchors, not controlled estimates that isolate a single determinant.

Question 1

Easy | SL/HL | 25 marks

Topic 2.2: law of supply and supply curve; Law: own price

(a) [10] Using an appropriate diagram, explain the relationship between the price of a good and its quantity supplied, ceteris paribus.

(b) [15] Using real-world examples, discuss whether higher prices always lead producers to offer more of a good for sale.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Show two points on the same upward-sloping supply curve; higher own price increases quantity supplied under unchanged conditions.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Show two points on the same upward-sloping supply curve; higher own price increases quantity supplied under unchanged conditions.

Counterargument and conditions. Observed sales and production can change with inputs, capacity and other determinants; a conditional relationship does not predict every observation.

Application and real-world example. USDA crop planting decisions (C) illustrate price incentives alongside other conditions.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 2

Medium | SL/HL | 25 marks

Topic 2.2: law of supply and supply curve; Law: quantity versus curve

(a) [10] Using an appropriate diagram, explain the difference between a decrease in supply and a decrease in quantity supplied.

(b) [15] Using real-world examples, evaluate whether lower output is sufficient evidence that supply has decreased.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. An own-price fall moves down along S; higher input costs shift the entire S left at a fixed own price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. An own-price fall moves down along S; higher input costs shift the entire S left at a fixed own price.

Counterargument and conditions. Output data alone cannot distinguish changed own price, costs, availability or other determinants.

Application and real-world example. ONS UK input-price increases in 2022 (B) offer context, not a controlled explanation of output.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 3

Hard | SL/HL | 25 marks

Topic 2.2: law of supply and supply curve; Law: schedule

(a) [10] Explain how a producer's supply schedule can be represented by an upward-sloping supply curve.

(b) [15] Using real-world examples, discuss the usefulness of a supply curve in understanding a producer's decisions when conditions change.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Connect each hypothetical own price to quantity willing and able to offer in the same period; plot P vertical and Q horizontal.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Connect each hypothetical own price to quantity willing and able to offer in the same period; plot P vertical and Q horizontal.

Counterargument and conditions. The schedule holds other determinants fixed and says nothing by itself about actual market sales or price.

Application and real-world example. A bakery's weekly output at alternative selling prices; named changes in energy costs (B) illustrate the qualification.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 4

Medium | SL/HL | 25 marks

Topic 2.2: law of supply and supply curve; Law: offered versus sold

(a) [10] Explain why the quantity a firm is willing and able to supply at a given price may differ from the quantity it actually sells.

(b) [15] Using real-world examples, evaluate whether changes in sales reliably indicate changes in a firm's willingness to supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Supply is a conditional offer at a price; actual sales also depend on buyers and transactions, not just the schedule.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Supply is a conditional offer at a price; actual sales also depend on buyers and transactions, not just the schedule.

Counterargument and conditions. Do not solve market equilibrium; inventory and unsold goods illustrate why offers and realised sales differ.

Application and real-world example. A retailer's unsold goods are a valid documented example; ONS retail data alone (B) cannot identify a firm's supply curve.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 5

Medium | SL/HL | 25 marks

Topic 2.2: individual producer and market supply; Market: summation

(a) [10] Explain the relationship between an individual producer's supply and market supply.

(b) [15] Using real-world examples, discuss whether one firm's increased output will significantly increase market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. At each common price, add the quantities that all firms are willing and able to supply; one firm's contribution depends on its share.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct. If aggregating firms, horizontally sum their quantities at a common price, including zero offers if applicable.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. At each common price, add the quantities that all firms are willing and able to supply; one firm's contribution depends on its share.

Counterargument and conditions. Other producers may change offers in the opposite direction; a dominant firm differs from a small farm.

Application and real-world example. A small farm and an entire national crop market offer contrasting scales (C).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 6

Easy | SL/HL | 25 marks

Topic 2.2: individual producer and market supply; Market: two firms

(a) [10] Explain how the supply schedules of two firms can be combined to obtain a market supply schedule.

(b) [15] Using real-world examples, evaluate the importance of differences between firms when constructing market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Add quantities horizontally at each stated price, allowing zero if a firm supplies none there.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct. If aggregating firms, horizontally sum their quantities at a common price, including zero offers if applicable.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Add quantities horizontally at each stated price, allowing zero if a firm supplies none there.

Counterargument and conditions. Different input costs or technology mean firms need not have identical supply schedules.

Application and real-world example. Different US crop farms documented in USDA planting surveys (A) illustrate heterogeneity.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 7

Hard | SL/HL | 25 marks

Topic 2.2: individual producer and market supply; Market: one versus all

(a) [10] Using an appropriate diagram, explain how a cost increase affecting every firm differs from one affecting only one firm in its effect on market supply.

(b) [15] Using real-world examples, discuss whether an economy-wide rise in input prices affects all producers in the same way.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Both can shift market S left, but coverage and size differ; derive market curve from firms' schedules at a common price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling. If aggregating firms, horizontally sum their quantities at a common price, including zero offers if applicable.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Both can shift market S left, but coverage and size differ; derive market curve from firms' schedules at a common price.

Counterargument and conditions. Input mixes, contracts and technology vary, so equal price change does not imply equal shifts.

Application and real-world example. ONS 2022 input-price rise (B) is an economy-wide anchor, not evidence of identical firm impacts.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 8

Medium | SL/HL | 25 marks

Topic 2.2: individual producer and market supply; Market: entrant

(a) [10] Explain how the entry of a new firm can increase market supply while existing firms' supply curves are unchanged.

(b) [15] Using real-world examples, evaluate whether new producers always make a large contribution to market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. New producer adds offered quantities at each price, shifting market S right without changing incumbent schedules.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct. If aggregating firms, horizontally sum their quantities at a common price, including zero offers if applicable.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. New producer adds offered quantities at each price, shifting market S right without changing incumbent schedules.

Counterargument and conditions. Its productive capacity and costs determine the size; incumbent exits or other shifts can offset.

Application and real-world example. A new bakery in a town versus an additional producer in a national industry.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 9

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: costs of factors of production; Costs: wages

(a) [10] Using an appropriate diagram, explain how a rise in wages can affect the supply of a labour-intensive good.

(b) [15] Using real-world examples, discuss whether a rise in wages necessarily reduces the supply of every good.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Higher labour cost reduces quantities profitable to offer at each output price and shifts S left, ceteris paribus.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Higher labour cost reduces quantities profitable to offer at each output price and shifts S left, ceteris paribus.

Counterargument and conditions. Labour share differs, and improved productivity or lower other costs can offset.

Application and real-world example. A restaurant or garment producer with rising labour costs; verify specifics in a chosen case.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 10

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: costs of factors of production; Costs: energy

(a) [10] Using an appropriate diagram, explain how a fall in energy costs may affect the supply of an energy-intensive product.

(b) [15] Using real-world examples, evaluate the importance of energy costs in explaining changes in supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Lower energy input cost makes more production viable at each price, shifting S right.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Lower energy input cost makes more production viable at each price, shifting S right.

Counterargument and conditions. Energy use, contracts and simultaneous changes in other inputs determine importance.

Application and real-world example. ONS UK producer input-price data (B) provide a real setting; a fall is a separate conditional scenario.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 11

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: costs of factors of production; Costs: imported inputs

(a) [10] Explain how a rise in the cost of a production input can shift supply even though the price of the final good is unchanged.

(b) [15] Using real-world examples, discuss whether a firm's own selling price is the main determinant of the amount it is willing to supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Input cost changes per-unit profitability at a fixed output price; distinguish S shift from movement due to output price.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Input cost changes per-unit profitability at a fixed output price; distinguish S shift from movement due to output price.

Counterargument and conditions. Costs, technology and other determinants may be more important in a period; no exchange-rate theory needed.

Application and real-world example. A food manufacturer facing higher packaging cost, linked to ONS producer input series (B).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 12

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: costs of factors of production; Costs: divergent inputs

(a) [10] Using an appropriate diagram, explain how a rise in one input cost and a fall in another may affect supply.

(b) [15] Using real-world examples, evaluate whether information about one input price is sufficient to predict a change in supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. One input tends to shift S left and another right; net shift depends on their relative importance.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. One input tends to shift S left and another right; net shift depends on their relative importance.

Counterargument and conditions. Input substitution and intensity are context-specific; no mandatory quantitative cost model.

Application and real-world example. A bakery with dearer flour but cheaper energy; B anchors real input-price variation.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 13

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: costs of factors of production; Costs: firm heterogeneity

(a) [10] Explain why a rise in the price of one factor of production may affect the supplies of two firms differently.

(b) [15] Using real-world examples, discuss whether input price data alone can explain an observed change in market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Different production methods and factor shares yield different firm shifts, aggregated horizontally.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Different production methods and factor shares yield different firm shifts, aggregated horizontally.

Counterargument and conditions. Other inputs, technology, numbers of firms and expectations also matter.

Application and real-world example. Energy-intensive and less energy-intensive UK manufacturers during 2022 (B).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 14

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in competitive supply; Competitive: rival price rises

(a) [10] Using an appropriate diagram, explain how a rise in the price of a good in competitive supply can affect the supply of another good.

(b) [15] Using real-world examples, discuss whether a higher price for one crop always reduces the supply of another crop.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. Shared land or resources shift toward dearer crop, shifting focal crop S left at its unchanged price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Shared land or resources shift toward dearer crop, shifting focal crop S left at its unchanged price.

Counterargument and conditions. Crop rotation, land suitability and contracts can constrain switching; other determinants may offset.

Application and real-world example. USDA corn and soybean planting research (C), without equating acreage mechanically with actual output.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 15

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in competitive supply; Competitive: rival price falls

(a) [10] Using an appropriate diagram, explain how a fall in the price of one good in competitive supply affects the supply of an alternative good.

(b) [15] Using real-world examples, evaluate the extent to which alternative product prices influence producers' decisions.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. Resources may shift from less profitable rival to focal good, shifting focal S right.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Resources may shift from less profitable rival to focal good, shifting focal S right.

Counterargument and conditions. Feasibility of changing production and relative costs determine response.

Application and real-world example. USDA crop acreage research (C) compares competing land uses.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 16

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in competitive supply; Competitive: own versus rival

(a) [10] Explain why a rise in the price of wheat has a different diagrammatic effect on the supply of wheat and the supply of a competing crop.

(b) [15] Using real-world examples, discuss whether evidence of increased wheat planting proves that wheat's supply curve shifted right.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. Wheat own-price rise moves along wheat S, while competing crop's S may shift left when land moves.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Wheat own-price rise moves along wheat S, while competing crop's S may shift left when land moves.

Counterargument and conditions. Planting is a production choice, not by itself a complete wheat supply schedule; other factors may shift wheat S.

Application and real-world example. USDA wheat and other crop planted-acreage studies (C).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 17

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in competitive supply; Competitive: resource constraint

(a) [10] Explain why two goods are in competitive supply when producers use the same limited resource.

(b) [15] Using real-world examples, evaluate whether higher prices for an alternative product are sufficient to predict a fall in another product's supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. One use of land, equipment or labour displaces the other, so alternative's price is a non-own-price determinant.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. One use of land, equipment or labour displaces the other, so alternative's price is a non-own-price determinant.

Counterargument and conditions. Different land suitability, crop rotations or fixed contracts weaken switching.

Application and real-world example. USDA notes rice fields' special preparation makes switching harder (C).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 18

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in joint supply; Joint: basic

(a) [10] Using an appropriate diagram, explain how a rise in the price of one jointly supplied product can affect the supply of another.

(b) [15] Using real-world examples, discuss whether a change in the price of one good always changes the supply of its by-product.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. Greater production of one jointly produced good yields more of the other, shifting its S right at unchanged own price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Greater production of one jointly produced good yields more of the other, shifting its S right at unchanged own price.

Counterargument and conditions. Technical proportions, processing choices, storage and disposal can complicate the link.

Application and real-world example. Crude-oil refining yields multiple products; specify petrol and diesel as a contextual example.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 19

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in joint supply; Joint: price decline

(a) [10] Using an appropriate diagram, explain how a fall in the price of one joint product may affect the supply of the other.

(b) [15] Using real-world examples, evaluate whether joint production alone determines the size of a supply shift.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. Reduced output of first may produce less co-product, shifting its S left at a given own price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Reduced output of first may produce less co-product, shifting its S left at a given own price.

Counterargument and conditions. Output mix and the relative importance of both products affect quantity; one observed price is insufficient.

Application and real-world example. Dairy processing into butter and skimmed milk, as illustrated in the uploaded textbook.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 20

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: prices of related goods in joint supply; Joint: contrast

(a) [10] Explain the difference between joint supply and competitive supply in the way a higher price for one good affects supply of another.

(b) [15] Using real-world examples, discuss the importance of production relationships in predicting how a price change affects another good's supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Competitive supply diverts shared resources between alternatives; joint supply produces more than one output from a common process.

Required reasoning. A price rise in joint product tends to shift co-product S right; in a competitive use it tends to shift rival S left.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. A price rise in joint product tends to shift co-product S right; in a competitive use it tends to shift rival S left.

Counterargument and conditions. Correct classification depends on technical production process and feasible substitution of resources.

Application and real-world example. Refinery products versus corn/soybean land use (C) provide distinct cases.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 21

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: indirect taxes and subsidies; Policy: indirect tax introduced

(a) [10] Using an appropriate diagram, explain how an indirect tax on production can affect a good's supply.

(b) [15] Using real-world examples, discuss whether a higher indirect tax necessarily causes the supply of a taxed good to decrease.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. An indirect tax adds to producer costs; a per-unit producer subsidy reduces effective costs at a given market price.

Required reasoning. A per-unit cost increase tends to shift S left/up at a fixed producer-relevant output price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not treat a profit tax as the specified indirect tax or assume every support payment is per-unit.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. A per-unit cost increase tends to shift S left/up at a fixed producer-relevant output price.

Counterargument and conditions. Exemptions, reformulation and changing input costs can alter observed supply; no demand or equilibrium incidence required.

Application and real-world example. UK Soft Drinks Industry Levy rates (D); distinguish taxed formulations and reformulated products.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 22

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: indirect taxes and subsidies; Policy: tax removal

(a) [10] Using an appropriate diagram, explain how removing an indirect tax can affect supply.

(b) [15] Using real-world examples, evaluate whether the removal of an indirect tax is sufficient to ensure an increase in supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. An indirect tax adds to producer costs; a per-unit producer subsidy reduces effective costs at a given market price.

Required reasoning. Lower per-unit cost tends to shift S right/down, ceteris paribus.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not treat a profit tax as the specified indirect tax or assume every support payment is per-unit.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Lower per-unit cost tends to shift S right/down, ceteris paribus.

Counterargument and conditions. Other costs, firms' ability to produce, and offsetting changes can prevent a net rise.

Application and real-world example. D's sugar-content thresholds help illustrate an exemption, not proof of output response.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 23

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: indirect taxes and subsidies; Policy: subsidy granted

(a) [10] Using an appropriate diagram, explain how a per-unit subsidy to producers can affect supply.

(b) [15] Using real-world examples, discuss whether a producer subsidy always leads to a larger quantity being offered at a given market price.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. An indirect tax adds to producer costs; a per-unit producer subsidy reduces effective costs at a given market price.

Required reasoning. Payment per unit lowers effective cost and shifts S right at the same stated market price.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not treat a profit tax as the specified indirect tax or assume every support payment is per-unit.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Payment per unit lowers effective cost and shifts S right at the same stated market price.

Counterargument and conditions. Eligibility, take-up, production constraints and concurrent costs matter; do not require welfare diagrams.

Application and real-world example. A documented agricultural producer payment is appropriate if its design is specified; avoid asserting all payments are per-unit.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 24

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: indirect taxes and subsidies; Policy: contrasting instruments

(a) [10] Explain why an indirect tax and a per-unit producer subsidy tend to shift supply in opposite directions.

(b) [15] Using real-world examples, evaluate whether the size of a tax or subsidy alone determines the change in supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. An indirect tax adds to producer costs; a per-unit producer subsidy reduces effective costs at a given market price.

Required reasoning. A tax raises net cost and shifts S left; a per-unit subsidy lowers effective cost and shifts S right.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not treat a profit tax as the specified indirect tax or assume every support payment is per-unit.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. A tax raises net cost and shifts S left; a per-unit subsidy lowers effective cost and shifts S right.

Counterargument and conditions. Different coverage, production methods and starting costs affect actual response; no calculation needed.

Application and real-world example. UK soft-drink levy (D) and a named unit-linked farm support scheme, if accurately described.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 25

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: future price expectations; Expectations: future rise

(a) [10] Using an appropriate diagram, explain how an expected future price rise may affect current supply of a storable good.

(b) [15] Using real-world examples, discuss whether producers should always withhold supply when they expect a higher future price.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Stock can be held for later sale, reducing current offered quantity at each current price; current S shifts left.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Stock can be held for later sale, reducing current offered quantity at each current price; current S shifts left.

Counterargument and conditions. Storage cost, perishability, cash needs and unreliable forecasts may favour present sale.

Application and real-world example. A grain warehouse with an observed expected-price announcement; USDA planting intentions (A) concern future production rather than withholding current stock.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 26

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: future price expectations; Expectations: future fall

(a) [10] Using an appropriate diagram, explain how an expected future price fall may affect current supply.

(b) [15] Using real-world examples, evaluate whether an anticipated future price fall must increase current supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. If feasible, producers bring sale forward and shift current S right.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. If feasible, producers bring sale forward and shift current S right.

Counterargument and conditions. Production timing, existing stock and contractual commitments can prevent acceleration.

Application and real-world example. A durable-goods producer selling existing inventory before an anticipated price decline.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 27

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: future price expectations; Expectations: current versus future

(a) [10] Explain how an actual rise in today's output price differs from an expected rise in a future output price in its effect on current supply.

(b) [15] Using real-world examples, discuss whether observed current output is enough to identify producers' price expectations.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Actual own-price rise is movement along S; expected future rise can shift current S left if output is withheld.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Actual own-price rise is movement along S; expected future rise can shift current S left if output is withheld.

Counterargument and conditions. Expectations are not directly observed, and future production incentives can differ from current stock decisions.

Application and real-world example. Grain-storage decisions compared with USDA planting-intention survey (A).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 28

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: future price expectations; Expectations: reversal

(a) [10] Explain how a change from expecting a future price rise to expecting a future price fall can affect current supply.

(b) [15] Using real-world examples, evaluate how useful producer expectations are in explaining shifts in supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Reduced incentive to delay sale can increase today's supply at given current price.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Reduced incentive to delay sale can increase today's supply at given current price.

Counterargument and conditions. Credibility, storage and alternative production choices qualify; other cost shocks may dominate.

Application and real-world example. USDA intended planting survey (A) shows plans, not a direct measurement of current marketed supply.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 29

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: technology; Technology: productivity

(a) [10] Using an appropriate diagram, explain how a productivity-improving technology affects supply.

(b) [15] Using real-world examples, discuss whether adopting new technology inevitably increases a firm's supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. More output from given resources or lower unit cost shifts S right at each output price, ceteris paribus.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. More output from given resources or lower unit cost shifts S right at each output price, ceteris paribus.

Counterargument and conditions. Installation, training and access costs can delay or offset; adopted machinery alone is no proof of greater marketed output.

Application and real-world example. Precision farming or more efficient manufacturing equipment, if supported by a named case.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 30

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: technology; Technology: diffusion

(a) [10] Explain why a technological improvement adopted by one firm may affect market supply less than one adopted by all firms.

(b) [15] Using real-world examples, evaluate the importance of how widely technology is adopted for market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Aggregate individual S schedules: one firm's rightward shift versus many firms' shifts.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Aggregate individual S schedules: one firm's rightward shift versus many firms' shifts.

Counterargument and conditions. Market size and adopting firms' output shares matter; technology may suit only some producers.

Application and real-world example. Large farm versus industry-wide precision farming provides contrasting scenarios.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 31

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: technology; Technology: cost and price

(a) [10] Using an appropriate diagram, explain why lower production costs arising from better technology shift supply rather than cause a movement along it.

(b) [15] Using real-world examples, discuss whether rising output after technological change establishes that technology caused a supply shift.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Technology is non-price determinant of focal good; at fixed own price more can be supplied.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Technology is non-price determinant of focal good; at fixed own price more can be supplied.

Counterargument and conditions. Own price, firm entry and input costs may also change; identify evidence of adoption and cost effect.

Application and real-world example. A manufacturer's documented automation rollout; ONS producer data (B) illustrate potential confounders.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 32

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: technology; Technology: unequal benefits

(a) [10] Explain why the same new production technology may shift the supply curves of two firms by different amounts.

(b) [15] Using real-world examples, evaluate whether technological progress is the most important influence on supply in every industry.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Different initial methods, ability to use and fixed adoption cost change per-unit benefits.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Different initial methods, ability to use and fixed adoption cost change per-unit benefits.

Counterargument and conditions. In some sectors input-price shocks or changes in firm numbers matter more.

Application and real-world example. Contrast mechanised crops with an energy-intensive processing plant (B/C) if technically justified.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 33

Hard | SL/HL | 25 marks

Topic 2.2: non-price determinant: number of firms; Firms: entry

(a) [10] Using an appropriate diagram, explain how an increase in the number of firms affects market supply.

(b) [15] Using real-world examples, discuss whether entry into a market necessarily increases the amount actually sold.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. New firms add supply at each price, rightward market S, ceteris paribus.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. New firms add supply at each price, rightward market S, ceteris paribus.

Counterargument and conditions. Demand, output price and incumbent responses influence sales; do not derive new equilibrium.

Application and real-world example. A local food market with new vendors is a case if dates and entrants are identified.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 34

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: number of firms; Firms: exit

(a) [10] Using an appropriate diagram, explain how firms leaving an industry affect market supply.

(b) [15] Using real-world examples, evaluate whether a reduction in the number of firms must reduce market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Removal of offered quantities shifts aggregate S left if incumbents unchanged.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Removal of offered quantities shifts aggregate S left if incumbents unchanged.

Counterargument and conditions. Remaining firms' capacity or technology may compensate; firm count alone gives no shift size.

Application and real-world example. Store or bakery closures in a named locality, with evidence of replacement supply.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 35

Medium | SL/HL | 25 marks

Topic 2.2: non-price determinant: number of firms; Firms: consolidation

(a) [10] Explain how market supply might remain unchanged even though the number of firms falls.

(b) [15] Using real-world examples, discuss whether counting firms is a reliable measure of supply capacity.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Remaining producers could expand their individual offers at each price enough to offset exits.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Remaining producers could expand their individual offers at each price enough to offset exits.

Counterargument and conditions. If they cannot, S falls; capacity, input access and technology distinguish cases.

Application and real-world example. A merger or consolidation in a local production industry, assessed with output evidence.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 36

Easy | SL/HL | 25 marks

Topic 2.2: non-price determinant: number of firms; Firms: entry versus costs

(a) [10] Using an appropriate diagram, explain how entry of firms and rising input costs may affect market supply simultaneously.

(b) [15] Using real-world examples, evaluate whether a greater number of producers guarantees an increase in market supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Entry tends right while costs shift participating firms' schedules left; net ambiguous.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Entry tends right while costs shift participating firms' schedules left; net ambiguous.

Counterargument and conditions. Magnitude and incidence matter; aggregate consistently at a common price.

Application and real-world example. ONS 2022 cost changes (B) with a documented entrant in an industry.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 37

Hard | SL/HL | 25 marks

Topic 2.2: movements along and shifts of supply; Mixed: own price and input cost

(a) [10] Using an appropriate diagram, explain how an increase in a good's price and an increase in an input cost can affect the quantity supplied.

(b) [15] Using real-world examples, discuss whether a higher output price together with lower production proves that the law of supply has failed.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Own price moves up/right along a given S; input cost shifts S left, potentially more than offsetting observed quantity.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Own price moves up/right along a given S; input cost shifts S left, potentially more than offsetting observed quantity.

Counterargument and conditions. Two observations under different conditions do not test the ceteris paribus law directly.

Application and real-world example. UK producer price and input price series in 2022 (B).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 38

Medium | SL/HL | 25 marks

Topic 2.2: movements along and shifts of supply; Mixed: technology and tax

(a) [10] Using an appropriate diagram, explain how a technological improvement and a new indirect tax may affect supply of the same good.

(b) [15] Using real-world examples, evaluate whether the direction of a supply change can be predicted when production costs change in opposite ways.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Technology tends S right while tax tends S left; net depends on their per-unit impact.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Technology tends S right while tax tends S left; net depends on their per-unit impact.

Counterargument and conditions. Timing, coverage and adoption differ; do not assume exact cancellation.

Application and real-world example. A soft-drink producer improving processing amid a levy (D) is a scenario, not a factual measured outcome.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 39

Hard | SL/HL | 25 marks

Topic 2.2: movements along and shifts of supply; Mixed: same shift different cause

(a) [10] Explain how a fall in input costs and a rise in the number of firms can each increase market supply.

(b) [15] Using real-world examples, discuss whether a rightward shift in supply identifies the factor that caused it.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Lower costs shift individual curves right; entry adds new curves; both increase aggregate Q at a common price.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Lower costs shift individual curves right; entry adds new curves; both increase aggregate Q at a common price.

Counterargument and conditions. Price-quantity observations alone cannot identify costs versus entry; they may occur together.

Application and real-world example. A town's food producers facing cheaper energy and new entrants; B anchors input prices.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 40

Medium | SL/HL | 25 marks

Topic 2.2: movements along and shifts of supply; Mixed: time and stock

(a) [10] Explain how an expected future price increase and an improvement in technology can affect current supply in opposite directions.

(b) [15] Using real-world examples, evaluate whether higher productive capacity always increases the amount a producer offers for sale today.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price.

Required reasoning. Withholding storable stock tends S left today; improved technology tends S right at fixed price.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Withholding storable stock tends S left today; improved technology tends S right at fixed price.

Counterargument and conditions. Perishability, storage and use of technology determine net outcome; distinguish production from current sale.

Application and real-world example. Grain producer investing in better harvesting while holding stock; USDA intentions (A) do not establish withholding.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 41

Easy | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: diminishing returns

(a) [10] Explain how the law of diminishing marginal returns can contribute to an upward-sloping supply curve.

(b) [15] Using real-world examples, discuss whether diminishing marginal returns necessarily explain every observed increase in output price.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. With a fixed factor and additional variable input, marginal physical output eventually falls; greater output tends to require more variable input per extra unit.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. With a fixed factor and additional variable input, marginal physical output eventually falls; greater output tends to require more variable input per extra unit.

Counterargument and conditions. Short-run fixed-factor conditions matter; technology or factor changes can shift S.

Application and real-world example. A farm adding workers to fixed land is illustrative; USDA acreage (C) gives sector context but not proof of diminishing returns.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 42

Medium | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: increasing MC

(a) [10] Explain how increasing marginal costs can help account for the law of supply.

(b) [15] Using real-world examples, evaluate whether rising marginal costs always limit the amount producers are willing to supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. If extra units cost more, higher output price is needed to make additional output worthwhile, other things equal.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. If extra units cost more, higher output price is needed to make additional output worthwhile, other things equal.

Counterargument and conditions. Marginal cost need not rise over every range; technology and input prices can change the schedule.

Application and real-world example. Extra production shifts for a bakery or factory, without numerical MC calculation.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 43

Hard | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: returns to cost

(a) [10] Explain the relationship between diminishing marginal returns to a variable factor and increasing marginal costs.

(b) [15] Using real-world examples, discuss whether higher labour costs and diminishing marginal returns affect supply in the same way.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. With fixed other factors, declining marginal product means more variable input per extra unit, raising MC at unchanged input price; higher wage raises cost schedule.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. With fixed other factors, declining marginal product means more variable input per extra unit, raising MC at unchanged input price; higher wage raises cost schedule.

Counterargument and conditions. One changes marginal productivity along production schedule, the other factor price and supply curve; do not conflate.

Application and real-world example. Restaurant staffing with fixed kitchen space versus a rise in wages.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 44

Medium | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: fixed factor

(a) [10] Explain why diminishing marginal returns depend on holding at least one factor of production fixed.

(b) [15] Using real-world examples, evaluate whether an increase in factory size would remove diminishing marginal returns from all production decisions.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. Crowding of additional variable factor on fixed space can reduce marginal product; expanding fixed factor changes conditions.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Crowding of additional variable factor on fixed space can reduce marginal product; expanding fixed factor changes conditions.

Counterargument and conditions. New bottlenecks may remain, and increasing MC may arise for other reasons; avoid economies-of-scale theory.

Application and real-world example. A fixed bakery oven compared with adding a second oven as a conditional example.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 45

Medium | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: total versus marginal

(a) [10] Explain why total output may continue to rise while the marginal output of an additional worker falls.

(b) [15] Using real-world examples, discuss whether lower marginal output means a producer should stop employing additional workers immediately.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. Diminishing marginal returns mean each added worker adds less output, not negative output; rising MC follows under unchanged wage.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Diminishing marginal returns mean each added worker adds less output, not negative output; rising MC follows under unchanged wage.

Counterargument and conditions. Decision depends on the extra output's price relative to extra cost; no numerical optimisation required.

Application and real-world example. Fixed greenhouse or kitchen production example.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 46

Easy | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: price incentive and MC

(a) [10] Using an appropriate diagram, explain how a rise in a good's price can make an additional unit worthwhile when marginal cost increases with output.

(b) [15] Using real-world examples, evaluate whether a price rise must increase output when marginal costs are rising.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. Higher own price can cover higher cost of extra units; move up/right along S under fixed costs and technology.

Expected diagram. Supply diagram: P (price of the FOCAL product) on vertical axis, Q supplied on horizontal axis, initial upward-sloping S0. An own-price change is a movement between clearly labelled points on the SAME curve. A non-price factor shifts the WHOLE curve S0 to S1 left/up or right/down; compare quantities at the SAME focal-good price. For opposing factors show two directional shifts or an explicitly ambiguous net effect; label intermediate curves/points as appropriate. No demand curve, market equilibrium, tax wedge, welfare area, PES or numerical calculation is required. Credit equivalent correct labeling.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Higher own price can cover higher cost of extra units; move up/right along S under fixed costs and technology.

Counterargument and conditions. Capacity, production timing and other determinants can limit realised output; do not require a separate MC graph.

Application and real-world example. Farm or manufacturing output where expanding short-run production is feasible.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 47

Hard | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: varying cost

(a) [10] Explain how a rise in the price of a variable input may affect marginal costs and the supply curve.

(b) [15] Using real-world examples, discuss whether increasing marginal costs and an input-price rise are equivalent explanations of lower supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. At each output, dearer input raises marginal cost and shifts S left; intrinsic rise in MC as Q expands helps explain slope.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. At each output, dearer input raises marginal cost and shifts S left; intrinsic rise in MC as Q expands helps explain slope.

Counterargument and conditions. An observed input cost change and movement to higher Q are different mechanisms; effects can coexist.

Application and real-world example. ONS UK producer input price data (B).

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 48

Medium | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: technology versus returns

(a) [10] Explain how improved technology can change a producer's supply while diminishing marginal returns to a variable factor still operate.

(b) [15] Using real-world examples, evaluate whether technological improvement invalidates the law of diminishing marginal returns.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. Technology shifts production possibilities, raising marginal product at a given input mix; within a fixed-tech, fixed-factor situation returns can still diminish.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Technology shifts production possibilities, raising marginal product at a given input mix; within a fixed-tech, fixed-factor situation returns can still diminish.

Counterargument and conditions. The new supply can shift right though remain upward sloping; effect size depends on adoption and costs.

Application and real-world example. Precision farming on fixed acreage, contrasted with a new machine.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 49

Hard | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: heterogeneity

(a) [10] Explain why marginal costs may rise at different rates for two producers of the same product.

(b) [15] Using real-world examples, discuss whether the supply of an industry can be inferred from the marginal costs of a single producer.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. Different fixed factors, technology and input availability produce differing additional-unit costs; market S aggregates firms.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. Different fixed factors, technology and input availability produce differing additional-unit costs; market S aggregates firms.

Counterargument and conditions. Representative firm may help if conditions similar, but one schedule alone cannot determine aggregate.

Application and real-world example. Large versus small bakeries facing different oven capacity.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Question 50

Medium | HL | 25 marks

Topic 2.2: HL assumptions underlying the law of supply: diminishing marginal returns and increasing marginal costs; HL: qualified law

(a) [10] Explain how diminishing marginal returns and increasing marginal costs support the law of supply under stated assumptions.

(b) [15] Using real-world examples, discuss whether a change in observed output contrary to a price change refutes the law of supply.

Part (a) - Indicative answer and marking guidance

Essential concepts. Supply is quantities producers are willing and able to offer at alternative prices in a given period, ceteris paribus. Quantity supplied refers to a single price. Diminishing marginal returns occurs as successive units of variable input add less output with a fixed factor; marginal cost is cost of an additional unit of output.

Required reasoning. With fixed factor and unchanged variable-input price, diminishing marginal product raises MC; higher price supports greater Q offered along S.

Expected diagram. No compulsory diagram. If a diagram is offered, check P vertical, Q horizontal, upward S, correct movement/shift and the causal explanation. Do not assume illegible labels are correct.

Common misconceptions. Confusing own-price changes (movement along S) with non-price changes (shift of S); treating observed sales as the entire conditional supply curve; deriving price or sales from supply alone. Do not say declining marginal product means falling total output, or equate increasing marginal cost with rising overall input prices.

Acceptable alternatives. Accept any accurate equivalent mechanism and real case. Do not require later-topic equilibrium, tax incidence, price elasticity, producer surplus, externality or welfare analysis.

Part (b) - Indicative answer and marking guidance

Main argument. With fixed factor and unchanged variable-input price, diminishing marginal product raises MC; higher price supports greater Q offered along S.

Counterargument and conditions. Changes in inputs, technology or firm count shift S; observed output is not equivalent to quantity supplied at a fixed set of conditions.

Application and real-world example. UK 2022 input-price episode (B) as confounder; no causal claim about each firm.

Evaluation and judgment. A strong answer assesses the stated proposition under the conditions described, uses at least one accurate real-world case, weighs a competing explanation, and gives a reasoned conditional judgment. Do not infer a unique market price or actual sales from the supply curve alone.

Source register

The uploaded Tragakes textbook, Economics for the IB Diploma, second edition, printed pp. 26-29, is a supplementary concept check. Other well-developed and accurate real cases should receive equal credit.

IB syllabus. IB Economics guide, first assessment 2022, printed p. 26 for Topic 2.2 and pp. 57-58 for Paper 1.

Open source

A. US Department of Agriculture, National Agricultural Statistics Service, Prospective Plantings, 28 March 2024: surveyed corn/soybean acreage intentions; intentions are not realised output.

Open source

B. UK Office for National Statistics, Producer Price Index 2022 correction and historical series. Input costs rose in 2022; aggregate indices do not isolate individual firms' output decisions.

Open source

C. US Department of Agriculture, Economic Research Service, As U.S. Farmers Respond to Crop Price Changes, Trends in Planted Acreage Emerge (2024): comparative planting and resource-allocation evidence.

Open source

D. UK HM Revenue and Customs, Soft Drinks Industry Levy statistics background and references: rates, sugar-content thresholds and coverage; do not infer a price or output effect without more evidence.

Open source

Final checks

50 paired questions with [10] and [15]; 40 SL/HL and 10 HL; 10 Easy, 25 Medium and 15 Hard. Separate student and teacher PDFs. The current syllabus, not the older textbook numbering, determines assessable content.